PERIODIC TABLE & ELEMENT PROPERTIES REFERENCE SHEET

A High-Density, Exam-Ready Conceptual & Formula Reference for EAPCET Chemistry

1. Historical Foundations & Modern Periodic Law

- **Döbereiner's Law of Triads (1817):** For a triad of chemically similar elements (A, B, C), the atomic mass of B is approximately the arithmetic mean of A and C:
  Formula: *Atomic Mass(B) ≈ [Atomic Mass(A) + Atomic Mass(C)] / 2*. *Limitation:* Applicable only to a very limited number of elements.
- **Newlands' Law of Octaves (1865):** Elements ordered by atomic mass show a recurrence of properties in **every 8th element** (analogous to musical octaves). Valid strictly up to Calcium.
- **Mendeleev's Periodic Law (1869):** Properties of elements are periodic functions of their **atomic masses**. Mendeleev famously predicted *Eka-Silicon*, which was later identified as **Germanium (Ge)**.
- **Moseley's Modern Periodic Law (1913):** Plot of the square root of frequency (√ν) of emitted X-rays against atomic number (Z) yields a straight line:
  Formula: $\sqrt{\nu} = a(Z - b)$ (where a, b are constants). Hence, properties of elements are periodic functions of **atomic numbers (Z)**.

2. Table Architecture & IUPAC System (Z > 100)

- **Modern Structure:** Consists of **7 Periods** (horizontal rows) and **18 Groups** (vertical columns). Period 1 is the shortest (2 elements: H, He). Lanthanides (4f) and Actinides (5f) are positioned at the bottom in the **f-block**.
- **Block Assignment Rule:** Elements are assigned to block (*s, p, d, f*) matching the orbital receiving the **last valence electron**.
  Example: Config [Ar] $3d^6\ 4s^2$ has its last valence electron entering the 3d subshell → **d-block** (Group 8, Transition Metals).
- **Group 17 (Halogens):** Collectively includes Fluorine, Chlorine, Bromine, Iodine, and Astatine.

IUPAC Systematic Nomenclature Rules for Heavy Elements (Z > 100)

Digit	Root	Abbr.	Digit	Root	Abbr.	Systematic IUPAC Nomenclature Examples
0	nil	n	**5**	pent	p	**Z = 115:** un + un + pent + iium = **Ununpentium (Uup)** [Official: Moscovium (Mc)]
1	un	u	**6**	hex	h	**Z = 118:** un + un + oct + ium = **Ununoctium (Uuo)** [Official: Oganesson (Og)]
2	bi	b	**7**	sept	s	*Rules:* Digits written in order; end suffix is always **-ium**.
3	tri	t	**8**	oct	o	*Digit 1 Representation:* Represented strictly by root **un**.
4	quad	q	**9**	enn	e	*Digit 0 Representation:* Represented strictly by root **nil**.

3. Atomic and Ionic Radii Principles

- **Periodic Trends:** Radii **decrease left-to-right** across a period (Z_{eff} increases, pulling outer electron cloud in). Radii **increase down a group** (new quantum shell *n* added, shielding outer shell from the nucleus).
  Period 2 Radius Trend: **F < O < N < C** (radius increases leftward).
- **Isoelectronic Species:** Atoms/ions sharing identical electron configurations. Size decreases with higher nuclear charge Z:
  Formula: *Ionic Radius $\propto$ 1 / Nuclear Charge (Z)*
  Example: For Na^+ (Z=11), Mg^{2+} (Z=12), and Al^{3+} (Z=13), all containing 10 e^-: $\mathbf{Na^+ > Mg^{2+} > Al^{3+}}$.
- **Cation Contraction:** A cation is **always smaller** than its parent atom ($r_{cation} < r_{parent}$). Loss of electrons often sheds the outermost shell and increases the **effective nuclear charge per remaining electron**.
- **Anion Expansion:** An anion is **always larger** than its parent atom ($r_{anion} > r_{parent}$). Addition of electrons increases **inter-electronic repulsion** in the same shell, expanding the cloud under constant nuclear charge.

4. Ionisation Enthalpy (IE) Rules & Key Anomalies

• **Thermodynamic Definition:** Minimum energy required to remove an electron from a gaseous atom. Generally **increases across periods** (increasing Z_{eff}, smaller size) and **decreases down groups** (size and shielding increase).
• **Successive Enthalpy Rule:** $IE_1 < IE_2 < IE_3$. Removing an electron from a positive cation requires overcoming massive electrostatic attraction compared to a neutral atom.
• **Period 3 Extremes: Sodium (Na)** has the lowest IE_1 (easily loses $3s^1$ to reach octet). **Chlorine (Cl)** has the highest IE_1 (strong nuclear pull, smallest size in Period 3 excluding Argon).

> **CRITICAL IONISATION ENTHALPY SUBSHELL EXCEPTIONS (Be/B and N/O)**
> • **Beryllium vs. Boron (Be > B):** IE_1 of Be ($1s^2\ 2s^2$) is higher than B ($1s^2\ 2s^2\ 2p^1$) because Be has a stable, completely filled 2s subshell. Removing a 2s electron (high penetration, closer to nucleus) is harder than removing Boron's lone, poorly shielded 2p electron.
> • **Nitrogen vs. Oxygen (N > O):** IE_1 of N ($1s^2\ 2s^2\ 2p^3$) is higher than O ($1s^2\ 2s^2\ 2p^4$) because N possesses an extra-stable, symmetrically **half-filled $2p^3$ subshell**. In Oxygen, the pairing of two electrons in a single 2p orbital creates electron-electron repulsion, making the first electron removal significantly easier.

5. Electron Gain Enthalpy ($\Delta_{eg}H$) & Electronegativity (χ)

• **Electron Gain Enthalpy ($\Delta_{eg}H$):** Energy change when a gaseous atom gains an electron. Becomes **more negative across a period** (Z_{eff} increases) and **less negative down a group** (size increases).
Period 2 Trend (Negative Value): **N < O < F** (Nitrogen has stable half-filled $2p^3$ and is near zero/positive).
• **Noble Gas Exception:** Large **positive** $\Delta_{eg}H$ due to extremely stable fully filled shell ($ns^2\ np^6$). Incoming electron must enter an unstable higher energy shell (n+1), requiring energy input.
• **Electronegativity (χ):** Measures tendency of a bonded atom to attract the shared electron pair in a covalent bond (Pauling Scale). Increases across period, decreases down group. **Fluorine** is the most electronegative element (χ = 4.0).
Period 2 Trend: **C < N < O < F**. Electronegativity is **inversely related to metallic character**.

> **THE HALOGEN ANOMALY: CHLORINE vs. FLUORINE ($\Delta_{eg}H$)**
> • **Chlorine (Cl) possesses the most negative electron gain enthalpy in the periodic table (Cl > F in negative magnitude)**. Although F is more electronegative, its extremely small, highly compact 2p subshell creates high **inter-electronic repulsion** among its 9 electrons. This repulsion offsets the nuclear attraction for an incoming electron. In Cl's larger 3p subshell, the added electron has plenty of space, resulting in minimal repulsion and a more exothermic release of energy ($\Delta_{eg}H_{Cl}$ = -349 kJ/mol vs. $\Delta_{eg}H_F$ = -328 kJ/mol).

6. Metallic Character, Nature of Oxides & Diagonal Relationships

• **Metallic Character:** Tendency to lose electrons easily (electropositivity). Decreases left-to-right, increases down a group.
• **Oxides Classification:** Metals on the left form **Basic Oxides** (react with water to form bases, e.g., $\mathbf{K_2O}$, Na_2O, MgO). Non-metals on the right form **Acidic Oxides** (react with water to form acids, e.g., SO_3, CO_2, SO_2, P_2O_5). Elements near metalloid boundary form **Amphoteric Oxides** (react with both acids and bases, e.g., $\mathbf{Al_2O_3}$, BeO).
Period 3 Acidity Trend: Na_2O (Strongly Basic) < MgO (Basic) < Al_2O_3 (Amphoteric) < SiO_2 < P_2O_5 < SO_3 < $\mathbf{Cl_2O_7}$ **(Most Acidic)**.
• **Diagonal Relationship (Li/Mg and Be/Al):** Similarities between Period 2 elements and the element diagonally to their right in Period 3 (e.g., **Li & Mg**, **Be & Al**). Caused by **comparable atomic/ionic radii** and **similar charge density / ionic potential (φ)**: *φ = Charge / Size*.
• **Anomalous Behavior of First Elements:** The first element of Groups 1, 2, 13-17 (e.g., Li, Be, B, C, N, O, F) differs from its group due to **unusually small size**, high electronegativity/IE, and **lack of vacant valence d-orbitals** (limiting valency to 4).

7. Earth Crust Abundance & Liquid Metals

• **Elemental Abundance: Oxygen (O)** is the most abundant element in the Earth's crust by mass (followed by Silicon).
• **Liquid Transition Metals: Mercury (Hg)** is highly anomalous among metals as it is a **liquid at room temperature**, owing to its unique filled-shell relativistic configuration which resists metallic lattice bonding and leads to an exceptionally low melting point.

Summary of Group and Period trends (left-to-right and top-to-bottom)

Property	Trend Across Period (L → R)	Trend Down Group (T → B)	Fundamental Underlying Mechanism
Atomic & Ionic Size	Decreases	Increases	Z_{eff} increases across; new shell *n* added down group.
Ionisation Enthalpy	Increases (excluding Be/B, N/O)	Decreases	Increased Z_{eff} binds valence e^-; shell shielding reduces pull down group.
Electron Gain Enthalpy	Becomes More Negative (excluding Noble Gases)	Becomes Less Negative (excluding Cl/F)	Z_{eff} increases electron pull; size expands down group reducing pull.
Electronegativity	Increases	Decreases	Atomic size decreases L-to-R; atomic size increases down a group.
Metallic Character	Decreases	Increases	IE increases across (harder to lose e^-); IE decreases down (easier to lose e^-).
Oxide Nature	Basic → Amphoteric → Acidic	Acidic → Amphoteric → Basic	Left-side metals form basic ionic oxides; right non-metals form acidic covalent.